

# Raising the Luminous Efficacy of LEDs

The possibility of the luminous efficacy to approach 200 lm/W in the future is not ruled out. While some companies are still deliberating on different factors to raise efficacy, Cree claims to have already achieved 200 lm/W in its labs in 2011!

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**L**uminous efficacy is said to be a measure of how well any source of light produces visible light. Component makers, specifically LED makers in Asia, are paying attention to pursuing the goal of raising the luminous efficacy of LEDs.

According to a spokesperson from Everlight Electronics based in Taiwan, it is mainly a matter of chip technologies. Different-sized chips paired with certain packages produce certain efficiencies. New chip technologies such as

flip chip, and blue and red HV combinations are also being implemented for high efficiencies.

Mainland China-based Shenzhen Sofull LED Lighting Technology will adopt quality LED chips, improve the light transmittance, control the Vf value and upgrade the workmanship of the package.

OSRAM Opto Semiconductor's Betty Lee from Wuxi, China, adds that this could be done by improving the chip technology, phosphor conversion efficiency, thermal performance and other optimisation in package design.

It would be interesting to see how LEDs are addressing voltages radically different from that of main electricity, and adopting new packaging processes such as chips-on-board (COB). COB is a new packaging technology for the LED light engine. Multiple LED chips are packaged together as one lighting module. When lit up, it looks like a lighting panel.

According to Everlight, voltages have never been a main issue with LEDs because all lighting fixtures still need a form of power supply to transform and convert. However, high-voltage LEDs allow the process of

transforming and conversion to be simplified, resulting in a more simple and efficient design. In this respect, COB is said to be a good, high-voltage platform.

Sofull adds that the addressing voltage consumes voltage when the main voltage makes the COB work. So it will be better if the addressing voltage is kept as low as possible to decrease the consumed power.

According to the OSRAM spokesperson, LEDs address voltage depending on the driver design. Some customers may look for lower forward voltage so as to maximise the number of LEDs connected in series, based on a constant-current driver solution, while others may look for high-forward-voltage LEDs to increase the overall Vf to come closer to the AC supply. The advantage is to maximise the conversion efficiency of the driver. For the COB, customers are said to be targeting the Vf of lower than 40V.

## Breakthroughs in smart LED lighting systems

There have been breakthrough developments in capacitors, heat dissipation and smart LED lighting systems.

According to the Everlight spokesperson, many different plastic heat-dissipation materials are being developed for certain applications. Specifically, the application would be higher-wattage, omnidirectional LED replacement bulbs such as 75W and 100W bulbs. As these applications must withstand greater wattages and heat within a small lamp, both the LED and heat-sinking materials need to be very efficient and cost-effective. This is



DURIS E 3 from OSRAM Opto Semiconductors. The light has the same appearance as a continuous strip of light (Image courtesy: [www.osram-os.com](http://www.osram-os.com))